

CE 398

COURSE SYLLABUS

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PURDUE UNIVERSITY
LYLES SCHOOL OF CIVIL ENGINEERING
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FALL 2025

SYLLABUS for CE 398 (Introduction to Civil Engineering Systems)

This syllabus provides indication of the outcomes of students efforts in meeting the course expectations. It also includes a summary of topics to be covered in the course.

Course Information

Course number and title: CE 39800 Introduction to Civil Engineering Systems Design
Class Location: ME 1061
Meeting day(s) and time(s). Tuesday and Thursdays, 9:00AM to 10:15AM
Instructional Modality (Face-to-Face)
Course credit hours: 3. Prerequisites (if any): N/A

Instructor(s) Contact Information

You may please reach the instructor via Brightspace, Purdue email, phone, text, or at the office.
Name of the main instructor: Samuel Labi Other instructors: Hosein Pourgholamali
Office Location: HAMP 1268, or G175A. Purdue Email Address: labi@purdue.edu
No fixed office hours: open door policy; or email me to let me know when you want to meet.
Student Consultation hours, times, and location: TBA.

Teaching Assistant

Name: TBD Office: HAMP 2155 and HAMP 1268; 494-7381/2206
Student Consultation hours, times, and location: TBD

Course Description

CE 398, a core course for undergraduate studies in Civil Engineering (CE), is useful to all undergraduate students intending to pursue a career in civil engineering and related disciplines. The course introduces a fundamental systems approach to address various tasks at the various phases of civil infrastructure systems development. The course provides a solid context and foundation in this regard, and addressing the following:

- The evolution of civil engineering disciplines, systems thinking, and the typical goals and objectives of civil engineering infrastructure systems.
- The various phases of CE infrastructure systems development and the tasks typically faced by civil engineers at each phase.
- The various systems-related analytical tools including optimization, simulation, economic analysis and network theory.
- Current and emerging issues associated with civil infrastructure systems development such as CE systems vulnerability, resilience, sustainability, automation of functions, connectivity of system infrastructure and users, smart cities, the internet of things, and big data.

The course exposes the CE undergraduate to real life instances and applications and lessons learned that are associated with each phase of the CE infrastructure system development process, and inculcates in the student, the need for life-long learning in today's fast-evolving world. The course also helps the CE undergraduate to develop vital skills of technical report-writing and oral presentation. The course applies the concepts to problems in the various areas of civil engineering and provides illustrations, examples, numerical problems, case histories, and case studies in the various civil engineering disciplines: architectural, construction, environmental, geotechnical, hydraulics, materials, structures, and transportation. The didactic mechanisms in the course includes lecture presentations, in-class exercises, video presentations, software demonstrations, a jeopardy competition, a term project, and lectures by external speakers.

Learning Resources, Technology & Texts

The textbook for the course is:

Introduction to Civil Engineering Systems, by S. Labi, published by Wiley, 2014.

Supplementary texts are:

Systems Engineering with Economics, Probability, Statistics, C.J. Khisty, J. Mohammadi, A. Amekudzi

Civil and Environmental Systems Engineering, by C.S. Revelle and E. Whitlatch

Relevant databases, software packages, web resources, hardware requirements (e.g., webcam for exam proctoring), and tutoring support will be announced and provided during the semester.

Also, supplemental course materials including powerpoints will be provided on the course "BRIGHTSPACE" web site.

Learning Outcomes

1. Provide the student with an **overall sequence through which a civil engineering infrastructure system is developed**, from needs assessment, to planning, design, construction, operations, maintenance, and end-of-life. Include infrastructure related to all fields of civil engineering (structures, geotechnical, transportation, hydraulic, architectural, environmental, etc.).

2. Expose the student to **basic concepts in systems engineering**.

3. Learn the various **analytical tasks related to systems-engineering** that civil engineers encounter at each phase of the CE system development process.

3. Equip the student with the requisite **analytical tools related to systems-engineering** needed to carry out the tasks that Civil Engineers encounter at the various phases of the CE system development process. This includes a distinction between the tools related to domain knowledge (which is learned in other CE courses) and systems-related knowledge (which are learned in this course. The latter include engineering economics, life-cycle costing, financial analysis, modeling, simulation, optimization, risk & reliability analysis, network analysis, and so on).

4. Help the student to design a CE system, component, or process to meet needs within constraints associated with economic, social, political, ethical, health & safety, constructability and sustainability criteria. This is done through a term project. The student is thus exposed to real-life instances and applications of key criteria for evaluation and decision making in various contexts that are associated with each phase of CE systems development.

5. Expose the student to **current and emerging issues** related to CE systems, such as smart cities, the internet of things, big data, infrastructure vulnerability, and system resilience, and environmental sustainability.

6. Help the student to prepare for **professional licensure** by exposing them to various topics relevant to the practice including equity, ethics and applications of ICT in CE systems development.

7. Provide the student an opportunity to develop vital management-related skills of technical report writing, team work, and communication, and research project execution.

8. Serve as an **integrated part of Purdue's strategic curriculum** that prepares undergraduate students for effective careers in CE.

Assignments and Homework Policy and Grade Distribution

There will be a mid-term exam. With or without prior notification, in-class exercises may be given.

The grading distribution for the course is as follows:

In-class exercises -- 30%, Mid-Term Exam -- 40%, Term Project -- 30%

Grading Scale

Grade Limits will be as follows:

90 – 100%	A
80 – 89.99%	B
70 – 79.99%	C
60 – 69.99%	D
<60%	F

In the course of the semester, the above grading scheme may be modified at the discretion of the Instructor. For further information, see University Regulations on Grades and Grade Reports at *https://catalog.purdue.edu/content.php?catoid=18&navoid=23275#grades-and-grade-reports*

Attendance Policy

You are expected to attend each lecture, except those that are designated as "optional". Absences should be preceded by notification (e-mail or otherwise).

As part of Purdue's return to normal operations as much as possible this semester, attendance may be a factor in course grades as outlined in University Senate academic regulations listed on the Academic Regulations & Student Conduct section of the University Catalog website (*https://catalog.purdue.edu/content.php?catoid=18&navoid=23275#classes*). The regulations provide excused absence guidance for cases of grief/bereavement, military service, jury duty, and parenting leave. In all other cases, the concluding language sets the following general expectation:

> "The University expects that students will attend classes for which they are registered. At times, however, either anticipated or unanticipated absences can occur. The student bears the responsibility of informing the instructor in a timely fashion, when possible. The instructor bears the responsibility of trying to accommodate the student either by excusing the student or allowing the student to make up work, when possible. The University expects both students and their instructors to approach problems with class attendance in a manner that is reasonable."

Students are responsible for class-related work missed because of an unavoidable absence. Further, make-up work is "at the discretion of the instructor."

- Physical presence in the classroom is not the same as engagement. All students are encouraged to engage actively in the class.
- Students should work with the instructor to remove barriers (such as lapses in off-campus internet connectivity) to an equitable learning experience.
- This course will have a robust presence in the Brightspace learning management system (LMS). Purdue has provided guidance for classrooms, instructional labs, and experiential courses on Purdue's Innovative Learning website.
- Students are encouraged to work early and often with the Instructor to communicate expectations regarding attendance. All graded assessments and activities in this course will be tied to course learning outcomes.

Academic Guidance in the Event a Student is Quarantined/Isolated

CE 398 students who become quarantined or isolated will receive maximum support towards their academic progress. This will include completion of assignments and course requirements remotely. The instructor will accommodate the student either by excusing the student or allowing the student to make up work, when possible.

Course Schedule

Please see CE 398 Lecture Schedule on Brightspace.

Academic Integrity

You may be invited (or required) to work with other students in this course on an assignment. If you are in doubt about the propriety of working with another student, please consult the instructor involved. Otherwise, the Purdue University policy for academic dishonesty (below) applies.

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Part 5, Section III-B-2-a, Student Regulations] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972]

Purdue's guidance on Academic Integrity can be found at https://www.purdue.edu/odos/academic-integrity/.

If, in completing an assignment, you use the information, ideas, etc. from someone else, you must give credit to that source. Most often, a simple citation should suffice. See also https://www.purdue.edu/policies/academic-research-affairs/ia3.html.

In this class, AI may be used only where (a) the lesson is about AI and the student may need to turn in some work using AI tools, (b) the student seeks to analyze data collected as part of the term project, in which case the use of AI tools must be disclosed.

Nondiscrimination Statement

This course will be conducted in accord with Purdue's policy statement on nondiscrimination (https://www.purdue.edu/purdue/ea_eou_statement.php):

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies.

A link to Purdue's Nondiscrimination Policy Statement is included in the Brightspace template under University Policies. Please refer to the policy link in Brightspace.

Accessibility

Purdue University is committed to making learning experiences accessible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack. Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your

goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at evans240@purdue.edu.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday. Considering the significant disruptions caused by the current global crisis as it related to COVID-19, students may submit requests for emergency assistance from the Critical Needs Fund.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Emergency Preparedness resources is located on the Brightspace shell under University Policies. Your syllabus can outline what students should do in emergency situations. During emergencies, the Instructor will communicate with students via email regarding assignment submissions. Students should review the resources found on the Emergency Preparedness Resources for students. Students should keep their cell phones on to receive Purdue ALERT text messages.

AI Policy

Students are expected to adhere to the guidelines provided by instructors for academic work so that no student gains an unfair advantage. Using or attempting to use unauthorized materials, information, study aids, notes, or any other device in any academic exercise will not be tolerated. Unauthorized materials may include anything which or anyone who gives a student assistance that has not been approved by the instructor in advance."
(Source: https://www.purdue.edu/teaching-learning/instructors/ai.php)
In this class, AI may be used only where (a) the lesson is about AI and the student may need to turn in some work using AI tools, (b) the student seeks to analyze data collected as part of the term project, in which case the use of AI tools must be disclosed.



Disclaimer

The course-specific aspects of this syllabus are subject to change.

COURSE OUTLINE

SECTION 1 INTRODUCTION

Civil Engineering Disciplines and their Evolution
What is a System? System Phases, Tasks, and Tools
System Goals and Objectives

SECTION 2 THE TASKS AT EACH PHASE OF SYSTEMS DEVELOPMENT

The Tasks at Each Phase of Systems Development

SECTION 3 THE TOOLS NEEDED TO CARRY OUT THE TASKS

Statistical Modeling
Numerical Simulation
Optimization (Continuous I)
Optimization (Continuous II)
Optimization (Discrete)
Genetic Algorithms
Artificial Intelligence and Machine Learning Tools
Engineering Economics I
Engineering Economics II
Multiple Criteria Analysis
Network Analysis I
Network Analysis II

SECTION 4 OTHER TOPICS RELATED TO CIVIL SYSTEMS DEVELOPMENT

Ethics in CE systems development
ICT Applications in CE systems -- automation, connectivity, data science, smart cities, the Internet of Things
Threats, Exposure, and Resilience
Sustainable development of CE systems
Concluding lecture